Not All Spines Are Created Equal: How TotalSegmentator Fails on Lumbosacral Transitional Vertebrae

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Abstract

Wrong-level spine surgery remains a persistent never-event, with the Lumbosacral Transitional Vertebra (LSTV, 5–35% prevalence) as its largest anatomic driver. We audit TotalSegmentator—the most widely deployed CT segmentation system and default back-end of 3D Slicer—and show that on lumbarization cases its output is anatomically indistinguishable from a wrong-level surgical plan: lacking an L6 class, it shifts lumbar labels caudally by one level, with junction-DSC dropping 29 points on Any-LSTV. No existing CT benchmark surfaces this—not TotalSegmentator's own evaluation, not Li et al. [2025] (the per-class DSC frontier on the matched COLONOG split), and not VERIDAH (vertebra-*labeling* SOTA on a private CT cohort, operating on pre-localized crops with no pelvis). We contribute: (i) an LSTV-stratified evaluation protocol (per-class DSC, junction-DSC over a 40 mm L5/S1 window, per-class voxel confusion at the junction) translating the level-shift mechanism into quantifiable wrong-level surgical risk; (ii) CTSPINOPELVIC1K—1,153 CT volumes across 802 patients with unified spinopelvic masks, 33 LSTV-positive cases across a 6-way phenotype taxonomy, and radiologist Castellvi typing (I–IV) on all 33; (iii) the first publicly-released merge-based LSTV-handling dual-spinopelvic CT segmenter, exceeding TotalSegmentator on sacrum (TotalSegmentator=0.817 zero-shot) and uniquely delivering both spine and pelvis in a single forward pass with released weights; (iv) an empirical isolation showing VERIDAH's training-side L5/L6 merge eliminates the collision driving TotalSegmentator's level shift but exposes a residual L4/last_lumbar collision on count-style sacralization, motivating VERIDAH's sequence predictor as the necessary disambiguation component. CTSPINOPELVIC1K, the protocol, and 5-fold checkpoints are released publicly.

1 Introduction

Wrong-level spine surgery is a persistent never-event in operative spine care [Konin and Walz, 2010]. Despite intraoperative imaging and surgical-pause checklists, the most common cause—failure to correctly count vertebral levels in the presence of an LSTV—continues to account for the majority of reported wrong-level lumbar incidents [Carrino et al., 2011]. The mechanism is well documented: a surgeon counts lumbar bodies upward from the sacrum, finds five (because lumbarization presents a sixth lumbar body that has not been recognized), and labels the bottom body as L5 when it is anatomically L6. Surgery proceeds one level caudal to the intended target.

Modern radiology workflows increasingly route automated segmentation as an upstream substrate for clinical decision-making. The most widely deployed CT segmentation system in this role is TotalSegmentator [Wasserthal et al., 2023]: an nnU-Net trained on 1,204 routine CTs to delineate 104

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anatomic structures, the default segmentation back-end of 3D Slicer, with thousands of clinical and research installations. The original paper reports an aggregate test-set Dice of 0.943 across all 104 classes and qualitatively documents two spine failure modes affecting neighboring-vertebrae label confusion (1% of cases) and neighboring-rib confusion (12%). TotalSegmentator is neither the only public CT spine segmenter nor the per-class DSC frontier on the matched CTSpine1K + CTPelvic1K (COLONOG) split (§2); it is, however, the model surgeons and downstream tools consume. This paper concerns one anatomic class on which TotalSegmentator does not perform at its headline level, and the consequences when it does not.

The clinical claim. On lumbarized spines, TotalSegmentator produces an output pattern that is, when consumed by surgical planning pipelines or by surgeons confirming operative levels, *indistinguishable from a wrong-level surgical plan*. Its label vocabulary contains L1–L5 but no L6 class. On a lumbarized spine (six lumbar bodies plus sacrum), the model distributes its five lumbar labels across the six bodies by shifting caudally by one anatomic level: the body that is anatomically L6 receives the L5 label, the body that is anatomically L5 receives the L4 label, and so on. A surgeon targeting the disc between predicted-L4 and predicted-L5 lands on the anatomical L5–L6 disc, one level caudal to the intended L4–L5 target. TotalSegmentator reproduces the canonical LSTV-induced miscount automatically and silently.

The architectural alternative: VERIDAH. Recent work [Möller et al., 2026] addresses transitional vertebra labeling via a two-stage architecture: a 3D DenseNet169 classifier on per-vertebra crops (four heads: label, region, transition, visibility), followed by a constrained sequence predictor that imposes anatomical-count constraints (4–6 lumbar bodies; the predicted sequence may end in two consecutive L5 labels, the second interpreted as L6). The classifier is trained with L5/L6 (and T12/T13) labels merged for class-imbalance reasons; disambiguation is deferred to the sequence predictor. The architectural commitment is empirically validated (97.22% CT subject-level agreement) but has never been tested against direct semantic segmentation, and it is unknown whether a sufficiently expressive voxel-wise network trained at full resolution with the same merge requires the sequence predictor at all. We supply the missing test.

The evaluation gap. Neither TotalSegmentator, VerSe [Sekuboyina et al., 2021], CTSpine1K [Deng et al., 2025], nor CTPelvic1K [Liu et al., 2021] report LSTV-stratified performance. Earlier CT labelers explicitly excluded the failure mode [Sekuboyina et al., 2020]. VERIDAH reports anomaly performance on a private cohort without isolating semantic-segmentation contribution. The result is a measurement vacuum: a production model emits a confident, locally well-formed, caudally-shifted prediction on a 5–35%-prevalence congenital variant with no error signal at evaluation time.

Closing the gap requires hardware. Fitting both lumbar spine and pelvis at $0.75\times0.75\times0.80$ mm spacing and $256\times320\times320$ patch consumes $\sim$100 GB GPU memory under `nnUNetResEncUNetPlans_100G`, available on H200 (140 GB HBM3e) but not on the L40S/A100/H100 GPUs that bracket TotalSegmentator's training configuration. Neither VERIDAH nor SPINEPS train at this combined spinopelvic resolution.

Contributions.

(i) **Wrong-level surgical risk in TS, quantified.** An LSTV-stratified evaluation protocol (per-class DSC, junction-DSC over a 40 mm L5/S1 window, per-class voxel confusion at the junction) measures TS's systematic one-level caudal shift on lumbarization (L4=0.388, L5=0.291; 95%+ of GT-L6 voxels labeled L5 on representative cases, §4) and structural absence of L6 prediction across all evaluation cases—translating the level-shift mechanism into quantifiable wrong-level surgical risk.

(ii) **CTSpinoPelvic1K: a stratified spinopelvic benchmark.** 1,153 abdominal CT records across 802 patients, unified spinopelvic masks, 33 LSTV-positive patients across a 6-way subtype taxonomy (lumb $n = 14$; sacr_count $n = 9$; full sacralization $n = 6$; semi-sacralization $n = 2$; ambiguous $n = 2$), and a registration-free fusion pipeline that resolves a previously undocumented Y-axis coordinate-convention mismatch between CTSpine1K and CTPelvic1K. Both source datasets share the COLONOG cohort that underlies VerSe and CTSpine1K, but neither encodes pelvic context with LSTV stratification. Radiologist Castellvi typing (I–IV) on all 33 LSTV cases is released as an additional annotation layer.

(iii) **First publicly-released merge-based LSTV-handling dual-spinopelvic CT segmenter.** A 5-fold CV nnU-Net 3d_fullres baseline on H200 hardware adopting VERIDAH's training-side merge (last_lumbar = L5+L6) at the voxel level, with pelvic-class supervision in a single forward pass. We exceed TS on sacrum and uniquely deliver both spine and pelvis with released weights (Table 6); §2 details the reproducibility tiers separating openly-released from paper-only baselines.

(iv) **Empirical isolation: merge alone is insufficient.** The merge eliminates the L5/L6 collision driving TS's level shift but exposes a residual L4/last_lumbar collision on count-style sacralization. This residual collision empirically isolates the component of VERIDAH's success attributable to its sequence predictor—a direction VERIDAH's authors could not ablate.

2 Related Work

Wrong-level spine surgery and LSTV. [Konin and Walz, 2010, Carrino et al., 2011, Sekharappa et al., 2014] document LSTV-induced miscounting (5–35% prevalence) without addressing automated segmentation as a miscount-propagation pathway. Kwak et al. [2025] detect LSTV on plain radiographs (binary; different modality).

Transitional anatomy as one shift. The anatomical literature treats the thoracolumbar and lumbosacral variants as two ends of one homeotic shift of the column: a rib-less T12, a lumbar rib, a lumbarized S1 and a sacralized L5 are the same displacement read at different junctions [du Plessis et al., 2018, Poolman et al., 2023], and twelfth-rib morphology co-varies with lumbosacral status on CT [Nagata et al., 2025]. Stump ribs have been detected automatically at scale [Möller et al., 2025b], but no public segmentation benchmark labels both junctions on the same spines, which is what a shift needs to be tested against.

Production CT spine and pelvic segmentation. TotalSegmentator [Wasserthal et al., 2023], an nnU-Net trained on 1,204 CTs and the default 3D Slicer back-end, has no L6 class and no LSTV stratification; we audit it as the production system downstream pipelines actually consume, not as a per-class DSC frontier. Li et al. [2025] hold the published per-class DSC frontier on a COLONOG split (731 scans; 440/146/145) but release neither weights nor code (paper-only) and do not stratify by LSTV. Liu et al. [2025] release FracSegNet (PENGWIN+CTPelvic1K), a pelvic-only cascaded 3D nnU-Net providing sacrum and L/R hipbone; their qualitative note of sacralization degradation (their §4.3) corroborates the LSTV gap our benchmark surfaces. Table 6 compares all baselines.

Vertebra-labeling SOTA: VERIDAH. Möller et al. [2026] address transitional-vertebra *labeling* (distinct from segmentation): a four-head 3D DenseNet169 classifier on per-vertebra crops plus a constrained sequence predictor enforcing 4–6 lumbar bodies (allowing terminal double-L5 to encode L6). On a private CT cohort of 1,536 volumes VERIDAH reports 99.18% subject-level perfect labeling versus prior CT SOTA [Meng et al., 2023] 77.81%. Three properties matter for us: VERIDAH (i) requires upstream centroids (VerSe-derived models or SPINEPS [Möller et al., 2025a], itself MRI-only); (ii) excludes pelvis; (iii) validates on a private CT cohort. Their ablation (Table 5) shows the sequence predictor adds substantial value, but the comparison against direct semantic segmentation on a public LSTV-stratified split is missing—the gap our CTSPINOPELVIC1K-trained baseline (§7) fills.

Benchmarks and evaluation rigor. VerSe [Sekuboyina et al., 2021] (460 CT volumes) is the only public CT benchmark whose label vocabulary includes L6 (label 25) but it is unstratified by LSTV and lacks pelvic GT; earlier CT labelers explicitly excluded the failure mode [Sekuboyina et al., 2020]. Stratified safety-critical reporting [Maier-Hein et al., 2024] motivates §6.

3 Failure-Mode Analysis: TS on LSTV

We evaluate TotalSegmentator v2 (`total` task) zero-shot at full resolution on the full CTSPINOPELVIC1K matched cohort. TotalSegmentator was not trained on any CTSPINOPELVIC1K case, so every case is valid; this provides 1,152 of CTSPINOPELVIC1K's 1,153 records (one record

Table 1: CTSPINOPELVIC1K v10 composition. Counts are from the released volumes and manifest; LSTV grades in Table 4.

Property	Value
Records (one per patient)	802
prone / supine	377 / 425
Label identifiers (contiguous, bone and hardware)	68
vertebrae C1–L6 (VerSe ids 1–25), sacrum, T13, S1	28
hips, femora	4
ribs, thirteen per side	26
lumbar ribs (left, right)	2
hardware subtypes	7
Records with an L6	18
Records with a lumbar rib	16
Records with a stump twelfth rib (ratio < 0.33)	98
Four-lumbar records (fused segment carried as S1)	9
Instrumented records (hardware labelled)	11
Castellvi-graded records (two-reader consensus)	33
Ribs evaluable / residual offsets	5,749 / 2

Table 2: TS zero-shot stratified by LSTV phenotype on CTSPINOPELVIC1K, full-resolution mode. Junction DSC over a 40 mm L5/S1 axial window. L6 count: cases with any L6 voxel emitted (structurally zero by label vocabulary). †: junction analysis not computed for separate-mode records.

Subgroup	n	L1	L2	L3	L4	L5	Sac	Hips	Jxn DSC
All cases	801	0.911	0.906	0.913	0.914	0.913	0.817	0.929	0.818
Normal	768	0.921	0.918	0.929	0.936	0.938	0.817	0.929	0.824
Any LSTV	33	0.681	0.615	0.523	0.367	0.314	0.822	0.929	0.534
Lumbarization	14	0.878	0.853	0.690	0.388	0.291	0.833	0.928	**0.357**
Sacralization (count-style)	9	0.265	0.100	0.055	0.001	0.000	0.814	0.926	0.697
Semi-sacralization	2	0.951	0.951	0.951	0.955	0.956	0.824	0.937	†
Sacralization (full)	6	0.718	0.676	0.673	0.724	0.793	0.826	0.932	0.636
Ambiguous	2	0.952	0.951	0.882	0.678	0.374	0.771	0.932	†
L6 EMIT									0 / 1152

dropped due to TS inference failure) aggregated to 801 patients via patient-level merging, including 33 LSTV-positive patients. Full-resolution mode isolates the failure as a property of label vocabulary and training distribution rather than inference resolution.

Table 2 reports performance by phenotype. On normal anatomy ($n = 768$) TotalSegmentator performs at its published level. On Any-LSTV ($n = 33$) every metric drops; lumbarization collapses caudally (L1=0.878 $\rightarrow$ L4=0.388 $\rightarrow$ L5=0.291) with junction-DSC at 0.357. Count-style sacralization ($n = 9$) collapses uniformly across all lumbar classes (L1=0.265 down to L5=0.000). L6 predictions are structurally zero across all 1,152 cases.

The level-shift mechanism. The asymmetry between lumbarization (severe, caudally progressive) and count-style sacralization (uniform collapse) follows from a single fact: TotalSegmentator's output head has no L6 class. For 6-vertebra anatomies the model compresses 6 anatomic units into 5 labels by shifting label assignment caudally; the most rostral body remains L1 (escape into thoracic vertebrae), but every successive body is misaligned, with the final body labeled L5 when it is anatomically L6. For 4-vertebra anatomies the model emits standard L1–L5, overshooting the GT lumbar count and producing a $+1$-level offset uniformly. The level-shift mechanism is the segmentation analog of the surgical miscount [Konin and Walz, 2010].

4 Case Study: Wrong-Level Surgical Risk in TS Output

Aggregate metrics establish the population pattern; the case study below establishes the mechanism at the level a surgeon would consume the output. Token 215 is a lumbarization case (six anatomical lumbar bodies, separate spine and pelvic GT records merged at patient level). TS's output head emits

only L1–L5, so on a 6-vertebra anatomy the bottom-most body (anatomical L6) cannot receive its correct label. Table 3 reports per-class DSC: L1 and L2 are well-placed (0.954, 0.927), L3 is partially correct (0.594), and **L4, L5, and L6 all collapse to dice zero**—a clean signature of the level-shift, with no GT-L4 voxel correctly labeled, no GT-L5 voxel correctly labeled, and no L6 voxel emitted at all. The pelvic classes (sacrum, hips) remain at near-Normal accuracy, confirming that the failure is local to the lumbosacral count and not a general degradation of TS on this patient.

Table 3: Token 215 (lumbarization, 6 lumbar bodies): per-class TS zero-shot DSC. The L4–L5–L6 row is the empirical signature of the level-shift failure—all three collapse to dice zero because TS compresses 6 anatomical bodies into 5 labels. Pelvic classes are unaffected. † TS's output head has no L6 label.

Class	L1	L2	L3	L4	L5	L6	Sacrum	Hip_L	Hip_R
DSC	0.954	0.927	0.594	**0.000**	**0.000**	**0.000**†	0.817	0.922	0.926

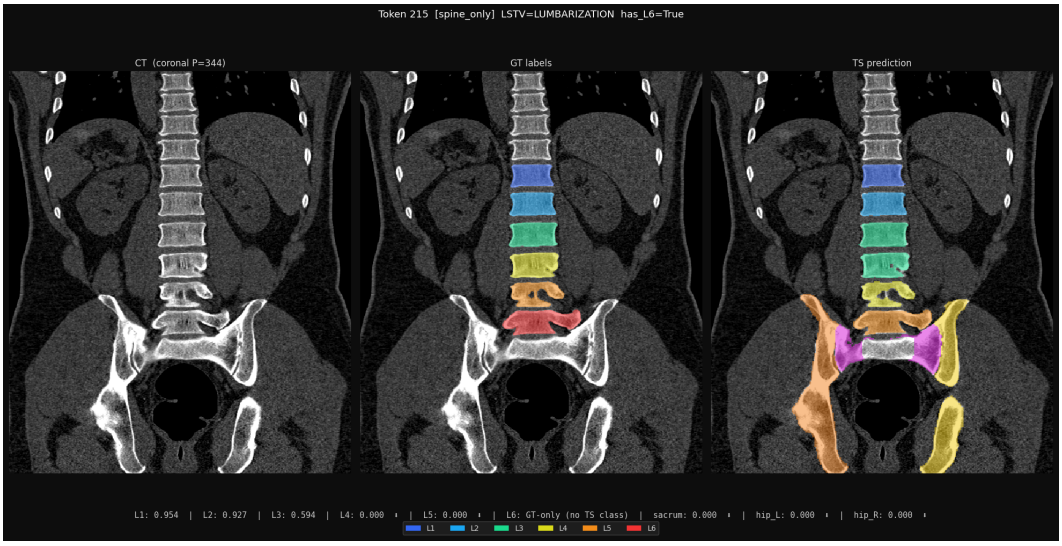


Figure 1: Token 215, lumbarization case (6 anatomical lumbar bodies). TS predictions overlaid on CT. The model compresses 6 bodies into 5 lumbar labels with a caudal cascade: the predicted-L3 region absorbs both anatomical L3 and L4 (yielding partial L3 dice 0.594, zero L4 dice); predicted-L4 sits over anatomical L5 (zero L5 dice); predicted-L5 sits over anatomical L6 (which has no label in TS's output head; zero L6 dice). The output is locally well-formed—five compact vertebral predictions in approximately correct positions—but every label below L2 is one anatomical level off, with the bottom body unlabeled-as-L6 entirely. Quantified in Table 3.

Surgical translation. A surgeon planning an L4–L5 microdiscectomy and using the segmentation to confirm operative level identifies the disc between predicted-L4 (anatomical L5) and predicted-L5 (anatomical L6). The targeted disc is L5–L6, one level caudal to the L4–L5 intent. The output is locally well-formed: five compact vertebral predictions in roughly correct positions. The error is detectable only by counting bodies (six lumbar, not five), which downstream pipelines do not perform.

5 CTSpinoPelvic1K: Construction and Composition

CTSpinoPelvic1K fuses CT scans from COLONOG [Smith et al., 2015] with annotations from CTSpine1K [Deng et al., 2025] (VerSe-derived per-vertebra masks) and CTPelvic1K [Liu et al., 2021] (sacrum and bilateral hips). The same COLONOG cohort underlies VerSe and CTSpine1K; CTSpinoPelvic1K's contribution is the patient-level fusion of CTSpine1K's vertebra masks with CTPelvic1K's pelvic masks, on one series per patient, plus the anatomy neither source labelled and the LSTV-stratification overlay neither provides. The release described here is v10 (September 2026):

802 records, one per patient, each on the patient's bone-richest acquisition; the 351 patients whose spine and pelvis were annotated on two different acquisitions are held out as a paired prone/supine resource rather than merged (Table 1).

What was added to the source annotations. Every record carries, beyond the CTSpine1K vertebrae and the CTPelvic1K sacrum and hips: per-level ribs on each side, a thirteenth-rib class per side so a rib borne by a T13 is not forced to be a twelfth, left and right lumbar ribs as their own classes so a rib borne by a lumbar body is not forced to be a twelfth either, both femora, a first sacral segment carved from the sacrum as the caudal anchor, and surgical hardware as its own class family (generic, cage, screw–rod, plate, arthroplasty, iliosacral screw, osteosynthesis). Lumbar levels are anchored on the lowest rib-bearing vertebra and on S1 rather than on a count, so a sixth lumbar body, a thirteenth thoracic body, a stump rib and a lumbar rib are recorded as what they are. The ribs began as model output (TotalSegmentator numbering completed by the Möller rib network) and were then reviewed by thirteen trained student annotators under a triaged protocol; the thoracic column added the same way. The release passes three validation layers: per-record geometric invariants (802 of 802), rib–vertebra incidence across 5,749 evaluable ribs (2 residual offsets, 0.035%), and agreement of derived spinopelvic measures with published reference values. The full build chain, from the public sources through every non-regenerable intermediate (source placements, the model-output base the students corrected, the review ledgers, the hand corrections, the radiologist hardware reads) to the released volumes, is archived separately (Zenodo 10.5281/zenodo.22647933) with a check that the final scripted hops regenerate the release voxel for voxel.

Patient-anchored fusion and coordinate correction. Patient identity is resolved via canonical UID matching (825/825 patients). Each mask is assigned to its source CT acquisition by intra-patient bone-HU maximization, $s^*(m) = \arg\max_{s \in \mathcal{S}_p} |\{v \in \mathcal{L}(m,s) : I_s(v) > 200\,\text{HU}\}| / |\mathcal{L}(m,s)|$, a metadata-free oracle since `SeriesDescription` fields are unreliable in TCIA releases (99.6% spine, 100% pelvic placement). CTSpine1K and CTPelvic1K were converted from DICOM with different `dcm2niix` invocations, producing a systematic Y-axis sign reflection in NIfTI affines; we detect via affine-derived center sign and reverse along axis 1 with affine update. The released volumes keep the VerSe-native vertebral identifiers (L6 = 25, so an L6 here is interoperable with VerSe and CTSpine1K) in a contiguous 68-identifier scheme; the merged last_lumbar scheme of §7 and the one-shot scheme of §8 are training-time remaps of that release, produced by one converter, never edits to it.

Six-way LSTV taxonomy with dual-provenance labels. We retain both LSTV label sources separately and partition into six subtypes by combining filename-qualifier evidence (CTPelvic1K morphological criterion) with vertebral-counting evidence (CTSpine1K VerSe-label-presence heuristic): **normal**, **lumb** (6 vertebrae labeled or VerSe label 25 present), **sacr_count** (4 vertebrae labeled with sacralization flag), **semisacralization**, **sacralization** (both via filename qualifier), and **ambiguous** (cross-source disagreement). An earlier substring-match parser bug collapsed semisacralization into full sacralization—merging two anatomically distinct phenotypes whose failure modes diverge sharply (full sacralization preserves the lumbar count and produces near-normal per-class DSC, whereas semisacralization introduces a partial transition that confuses the L5/S1 boundary). Any benchmark reporting an aggregate sacralization DSC under this collapse averages over two qualitatively different model behaviors; we identify and fix this. Without the corrected 6-way taxonomy, the qualitatively different failure patterns across LSTV variants (§3) are invisible.

Partial-annotation training protocol. Training only on fused records discards 70% of the data. We use nnU-Net v2's `ignore_label` [Isensee et al., 2024] to train on all 1,153: voxels in unsupervised regions are written as label 9, contributing zero to Dice and cross-entropy. Spine classes (1–5) receive supervision from 784 records; pelvic classes (6–8) from 714. This is a $3.36\times$ effective expansion. Per-record subtype refinement gates training-time tags on supervised anatomy.

Castellvi annotation layer. The 6-way taxonomy captures the *count* dimension of LSTV. The Castellvi classification [Castellvi et al., 1984] captures the orthogonal *morphological* axis: Type I (dysplastic transverse process), II (pseudoarticulation), III (bony fusion), or IV (mixed II/III), each `a` (unilateral) or `b` (bilateral). It is the surgeon's pre-operative wrong-level-risk indicator, and what a CT segmenter would need to predict to flag LSTV for review rather than silently emit a caudally-shifted label. Two resident radiologists jointly annotated all 33 LSTV cases by consensus (no

independent inter-rater scoring), extending the schema—originally sacralization-specific—to score the bony junction uniformly across variants, with per-side `a/b` laterality.

Table 4: Castellvi distribution across all 33 LSTV cases of CTSPINOPELVIC1K, cross-tabulated by 6-way count taxonomy.

Subtype	**Ia**	**Ib**	**IIa**	**IIb**	**IIIa**	**IIIb**	**IV**	**Total**
Lumbarization	0	1	1	3	1	7	1	14
Sacralization (count-style)	0	0	0	0	0	9	0	9
Sacralization (full)	0	0	0	0	1	3	2	6
Semi-sacralization	0	0	1	0	1	0	0	2
Ambiguous	0	0	0	1	0	0	1	2
Total	0	1	2	4	3	19	4	33

Morphology-stratified failure interpretation. The cross-tabulation (Table 4) surfaces two findings with direct bearing on model behavior. (1) **All 9 count-style sacralization cases are Castellvi IIIb**—bilateral bony fusion—and 19/33 (58%) of LSTV overall is IIIb. The residual L4/last_lumbar collision (§9.1) therefore occurs precisely when the discriminating cue is locally observable: bony fusion of the bottom lumbar to sacrum is visually distinct from a normal L5–S1 junction, yet the segmenter still emits last_lumbar for those voxels. Direct semantic segmentation has not learned the Castellvi correlate of the count anomaly even when it is morphologically present in the input. That is the empirical case for the one-shot arm of §8, which names the border bodies by type from their own appearance and leaves the count to a decoder, rather than for a global count constraint that would inherit the same ambiguity. (2) Only 1/33 cases is Type I (population prevalence 14–16% [Sekharappa et al., 2014]), reflecting source-dataset selection bias toward morphologically conspicuous presentations. Reported performance therefore characterizes the conspicuous regime; the clinically subtler Type I fraction is uncharacterized in this benchmark.

6 Evaluation Protocol

We define four metrics as the minimum required to attribute LSTV failure modes to wrong-level surgical risk [Maier-Hein et al., 2024]: (1) per-class DSC stratified by 6-way LSTV subgroup, exposing class-level degradation; (2) junction-DSC over a 40 mm L5/S1 axial window, the clinical window for surgical planning; (3) per-class voxel confusion at the L5/S1 junction, which exposes structured failure modes (the level shift) that DSC alone cannot characterize; and (4) for the merged-label baseline, three focused per-epoch confusion blocks anchored on the L4/last_lumbar boundary that drives the residual collision (`last_lumbar_GT_on_lumb`, `L4_GT_on_sacr_count`, `pred_last_lumbar_on_sacr_count`) plus a specificity number (`last_lumbar_specificity_on_sacr_count`).

The one-shot arm and the competitors are scored in one space. For the one-shot arm (§8) and for every public system we compare against, the unit of evaluation is the same: the 22-class one-shot labels of all 802 records, our own network on its validation fold, each competitor on all 802 (none was trained on them). Every competitor's native output is mapped into that space by class name, ribs one to eleven pooled per side, its sacrum and S1 classes unioned onto our whole sacrum, its femora pooled, and anything above T10 sent to background, where our labels are ignore in any case. Four numbers are then read off with the same code for every system: per-class Dice on typical and on transitional records separately; the rib-free count and its exact-match rate by transitional status; where the ground-truth twelfth-rib, thirteenth-rib and lumbar-rib voxels went; and whether the rare classes (L6, T13, the lumbar ribs) are emitted at all on the records that carry them. Hard-label systems pass through the sequence decoder as one-hot, so a system without an L6 class still has its *count* scored fairly, since two touching bodies with one name are split, while its *name* is scored as the miss it is. The systems: TotalSegmentator v2 (`total` task, full resolution, no region subset so that no second model crops the input) [Wasserthal et al., 2023]; SPINEPS-CT with the VERIDAH labeling head, the one competitor that can name an L6 and a T13 [Möller et al., 2025a, 2026], with Möller's rib network and rib-assignment code beside it [Möller et al., 2025b]; NV-Segment-CT, the released VISTA3D lineage; and RibSeg v2 for rib numbering. **[PENDING: numbers; runs launched 8 September 2026.]**

7 Spinopelvic Baseline

Architecture and training. nnU-Net v2 [Isensee et al., 2021, 2024] ResEnc-M planner, `3d_fullres`, `nnUNetResEncUNetPlans_100G` (100 GB GPU budget). Spacing $0.80\times0.75\times0.75$ mm, patch $\mathbf{256}\times\mathbf{320}\times\mathbf{320}$, batch 2; 7-stage residual encoder ($n_{\text{features}}=[32, 64, 128, 256, 320, 320, 320]$, $n_{\text{blocks}}=[1, 3, 4, 6, 6, 6, 6]$, 3^3 kernels, 2^3 strides, instance norm, deep supervision at 5 levels); 9-channel output (bg + 8 fg); ignore-label masked from loss. Class-reweighted DC_and_CE loss ($\mathbf{w}=[0.5, 1, 1, 1, 1, \mathbf{1.5}, \mathbf{2.0}, 1, 1]$; last_lumbar boost targets the clinically-important boundary, sacrum boost supports Castellvi precision); SGD Nesterov 0.99, $\lambda=3\times10^{-5}$, PolyLR 10^{-2}, 500 epochs (250 train + 50 val iters); 5-fold patient-stratified CV with 6-way LSTV stratification.

Two-tier LSTV emphasis. **Queue-level oversampling**: all five non-normal subtypes duplicated to 25% of per-fold training samples (matching VERIDAH's $4\times$ rate). **Class-reweighted CE** (above). No per-patch foreground biasing: under merged labels, biasing toward an L6 class is ill-defined, and biasing the L4/last_lumbar boundary on sacr_count would confound the residual-collision measurement.

Hardware. H200 NVL (140 GB HBM3e), 24 CPU workers, preunpacking once per job to a 1.7 TB local-NVMe cache (replacing NFS reads that throttled at 21 MB/s under 5-fold contention). Per-epoch $\approx$ 400–700 s; per-fold ≈ 3 days for 500 epochs. Reducing the GPU memory budget forces either smaller patch (loses spinopelvic context) or coarser spacing (loses Castellvi-defining detail); 100 GB is the smallest budget fitting both.

8 One-Shot Level Identity and the Sequence Decoder

The merged baseline of §7 closes the level-shift gap by construction: it refuses to name L5 and L6 apart, so it can never shift them. That is the right engineering answer for a segmenter that must not mislead a surgeon, and the wrong instrument for the scientific question the residual collision raises (§9.1): whether a vertebra's identity is *locally* decidable at all. The one-shot arm asks the network to name every level from its own appearance, with no count anywhere in the target, and then adds the one thing a semantic network cannot supply, the reasoning about the sequence.

The premise, tested before a GPU was spent. If T12 and L1 were not different shapes, a local classifier could only learn the annotators' convention. On the released morphometrics, eleven shape features per vertebra, each divided by the patient's own median across levels so that size is removed and only shape remains, separate thoracic from lumbar at AUC 0.998 and T12 from L1 at AUC 0.990 (accuracy 97.3% over 1,485 vertebrae, plain logistic regression, cross-validated by patient). Transverse-process span carries most of the weight; costal facets and facet-joint orientation are not in the feature set at all, and a network sees them directly.

Target. The release is remapped to 43 foreground classes plus ignore (Dataset813): T10, T11, T12 and T13 as distinct thoracic classes; L1–L6 as distinct lumbar classes; the sacrum; every rib named per side, one to thirteen, with the lumbar rib as its own class per side; the hips; the femora; and one derived class, the intervertebral space, computed at conversion from the per-level source labels wherever two consecutive bodies are both present. The lumbosacral disc is deliberately not derived, since whether one exists is the transitional finding. With the disc as a class, every vertebra is its own connected component of the vertebra classes, so an instance can carry one name only, by construction of the decoding rather than by hope. The twelfth rib and the lumbar rib are separate classes because that pair *is* the discrimination under test, and L5 is kept apart from L6 because whether the network can tell them apart is the question rather than a nuisance. An earlier arm with ribs one to eleven pooled (22 classes, Dataset810) was retired at epoch 35 once it had shown L6 learned as its own class (Fig. 5); the numbered-rib arm keeps every property of it and adds the ribs.

Sampler. nnU-Net's foreground oversampling picks a class inside an already-selected case, and case selection is uniform, so a class present in 16 of 802 records is seen a fraction of a percent of the time and is learned, correctly by its loss, as never to be predicted. Two changes address that: records carrying any variant (lumbosacral subtype, a lumbar rib, a T13) are drawn to half of every training

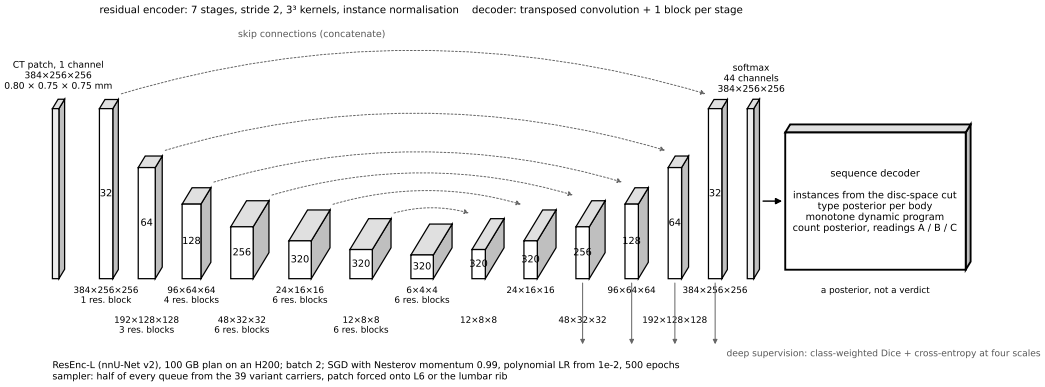


Figure 2: The network. A seven-stage residual encoder (channels 32 to 320, one to six residual blocks per stage) and a mirrored decoder with skip connections and deep supervision, a 44-way softmax, and the sequence decoder that turns voxels into a count with a posterior. Boxes are labelled with channels; spatial extents below.

queue, and inside such a record the patch is forced onto the rare class (the lumbar rib where present, else L6) with probability 0.6. Class weights in the cross-entropy term are 3.0 for L6 and the lumbar ribs, 2.5 for T13 and the thirteenth ribs, 2.0 for the twelfth ribs, 1.5 for T12, L5 and the sacrum, 1 elsewhere, 0.5 for background.

Architecture and training. nnU-Net v2 ResEnc-L, `3d_fullres`, 100 GB memory target (Fig. 2). The planner's patch, $256\times320\times320$ voxels at $0.80\times0.75\times0.75$ mm, spans 20.5 cm along the column; counting needs T10 to the sacrum, about 30 cm. The patch was therefore reshaped to $384\times256\times256$, 30.7 cm along the column and 19.3 cm in-plane, at the same voxel budget (25.2 against 26.2 million) and the same batch of two on an H200. The choice is measured, not argued (Fig. 3): over all 802 label volumes, a window of the planner's shape centred on the column holds 5.1 whole vertebral levels, 3.6% labelled voxels, and both counting anchors, the sacrum and a rib-bearing body, in 30% of scans; the column-shaped window holds 9.5 levels, 6.6% labelled voxels, and both anchors in 98%. Placed on the lumbosacral junction, where the sampler puts the rare-class patches, both shapes see the anchors (99%) but the column shape still holds nine whole levels to the planner's five. The price is the iliac wing, which the column shape clips; the hips are learned from the patches the sampler places on them. A longer shape, $448\times224\times224$, gains little more (9.6 levels) and cuts the pelvis harder. All three, and a $512\times192\times192$ extreme, train on one fold as an ablation; the shape reported here is the one the validation curves select **[PENDING]**.

The same five patient-grouped LSTV-stratified folds as the baseline, 500 epochs per fold, reading the preprocessed volumes from the shared filesystem after a single unpack. Validation predictions are written with their softmax probabilities.

Decoder. A semantic network can output "T12, L1, T12, L2" for four consecutive bodies. The decoder adds only the anatomical constraint and nothing learned. Vertebra instances are the connected components of each name class, with same-name fragments within a body height merged. For each instance the mean softmax over its voxels is folded into three type probabilities, thoracic (T10–T13), lumbar (L1–L6) and sacral, and a most likely name; rib classes within 4 mm are recorded. Ordering instances from superior to inferior, the boundary k between thoracic and lumbar is chosen to maximise the summed log type probability, and the softmax of that score over k is a posterior over the rib-free count. When the most likely count is six, the three readings of §1 are scored: B (a rib-less T12 counted as lumbar) is the posterior mass on $k{+}1$; C (a lumbarized S1) is the sacral share of the lowest instance's type mass; A is the remainder; and the double shift, both at once, is reported as their product rather than excluded. On one-hot ground truth the decoder reproduces the label's rib-free count on every record tested, which establishes the instance handling and the axis conventions before any network output exists.

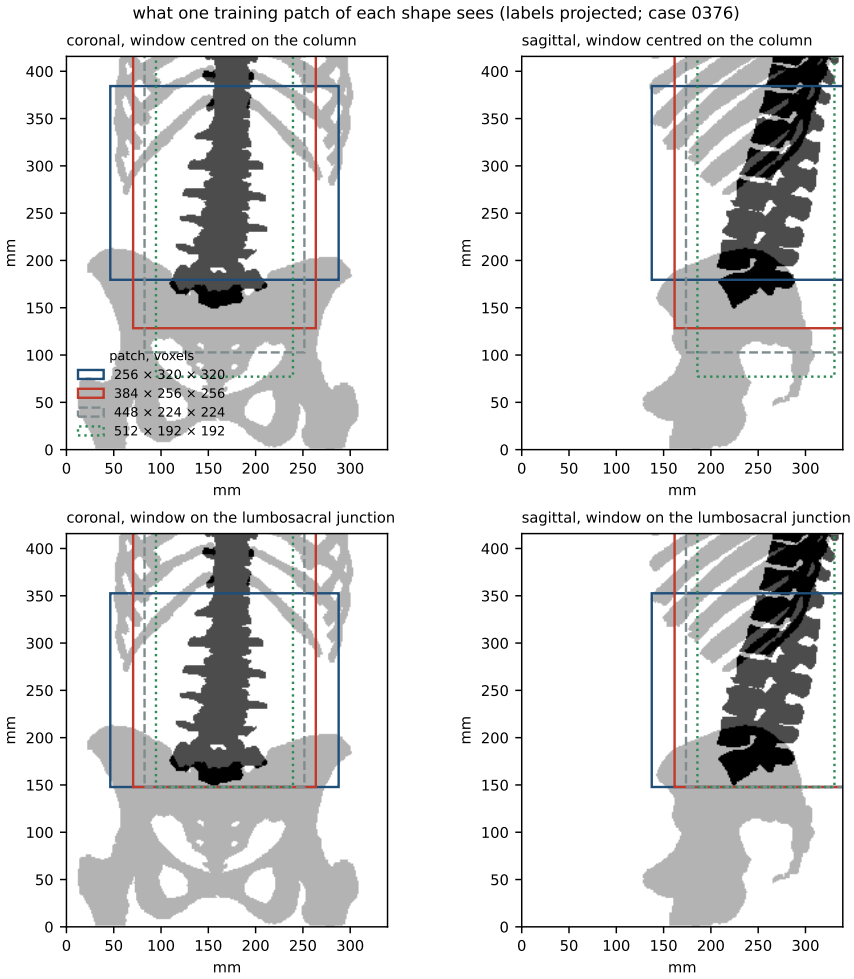


Figure 3: One training patch of each shape on a six-lumbar spine, labels projected coronally and sagittally, centred on the column (top) and on the lumbosacral junction (bottom). The planner's square patch ends at the sacral promontory; the column shapes reach from T10 through the sacrum and clip the iliac wings.

Every rib named. The released labels number thirteen ribs per side. The one-shot arm pools ribs one to eleven to spend its capacity at the junction; a second arm with the same sampler and every rib as its own class (42 foreground classes, Dataset813) is queued so that no public system numbers ribs where ours does not (Table 5). **[PENDING]**

What it cannot do. The names L5 and L6 are a count; the decoder does not pretend otherwise. What it decides is the *type* of each border body, and the posterior it reports is the honest output on a study that contains no C2.

9 Results

Headline. The merged-label spinopelvic baseline closes the LSTV junction-DSC gap on lumbarization: junction-DSC rises from TS = 0.357 to ours = **0.946** $\pm$ 0.007 (5-fold mean $\pm$ std; ensemble 0.950), a Δ= **+58.9** $\pm$ 0.7 points (5-fold; ensemble +59.3 points). The model emits non-degenerate last_lumbar on all 3/3 held-out lumbarization test cases with per-case last_lumbar DSC of 0.698 ± 0.038 (5-fold; ensemble 0.678), eliminating the structural absence that drives TS's level shift. A residual class collision persists on count-style sacralization: voxel-level confusion (§9.1, Block 2;

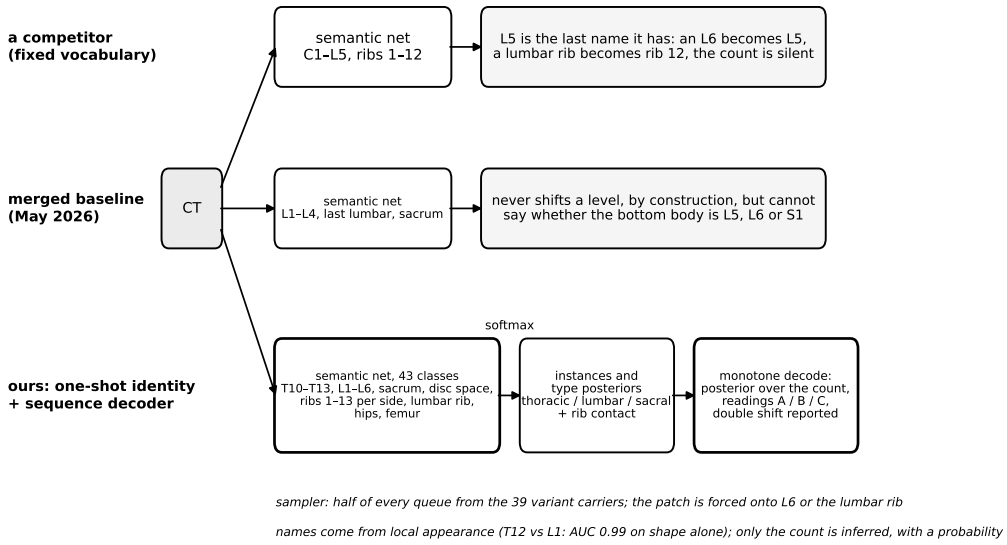


Figure 4: Three ways to read a column. A fixed-vocabulary competitor and our merged baseline each stop where their names stop. The one-shot arm names every level from its own appearance and hands the count to a decoder that returns a posterior.

Table 5: What each system can say about a spine. Filled: a class or output the released system produces. Open: in the released labels; the arm that names every rib per side (Dataset813) is queued behind the one-shot folds.

	TotalSeg. v2	NV-Segment-CT	SPINEPS+VERIDAH	RibSeg v2	**ours**
vertebrae T10–L5 by name	■	■	■		■
sacrum (and S1)	■	■	■		■
hips and femora	■	■			■
ribs, numbered per side	■	■		■	■
twelfth rib as its own class	■	■		■	■
a rib on L1 (lumbar rib)					■
a sixth lumbar body (L6)			■		■
a thirteenth thoracic body (T13)			■		■
voxel probabilities released					■
posterior over the rib-free count					■
readings A / B / C for six bodies					■
trained on labelled transitional anatomy			■		■
public CT test set with LSTV strata					■

$\sim$720K aggregated GT-L4 voxels across 5 folds) shows **82.7%** of GT-L4 voxels mis-predicted as last_lumbar, with per-case L4 DSC at 0.187 ± 0.337 (high cross-fold std driven by a single fold's outlier; the voxel-level analysis (Table 8) is unaffected because its denominator is dominated by the other folds). §9.1 characterizes this as the empirical signature of direct semantic segmentation's limit on transitional anatomy. On pelvic classes, our model exceeds TotalSegmentator on sacrum by $\Delta =$ **+15.2** $\pm$ 0.2 points (TS = $0.817 \rightarrow$ **0.969** $\pm$ 0.002 5-fold; 0.970 ensemble) and establishes state-of-the-art for openly-released dual spinopelvic CT segmentation (Table 6).

LSTV-stratified per-subgroup view. Table 7 exposes the LSTV-induced degradation that aggregate per-class DSC masks; the residual L4 DSC drop on sacr_count cases (§9.1) persists.

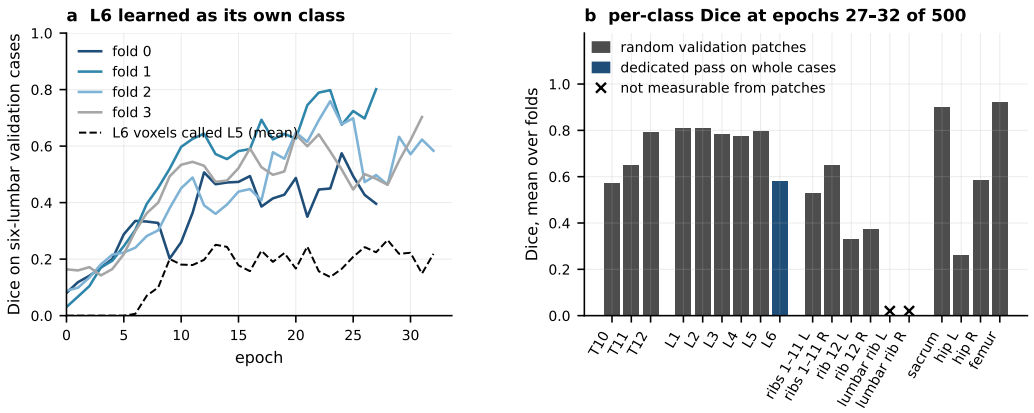


Figure 5: Training-time evidence, four folds, first thirty epochs of 500. (a) L6 Dice on the six-lumbar validation records from the dedicated pass, against the fraction of L6 voxels called L5. (b) Per-class Dice at the latest epoch; the lumbar ribs are not measurable from random validation patches.

Table 6: Per-class DSC ($\times 100$). **L5** is the unmerged anatomical L5 class; **last_lumbar** is the merged class L5+L6 (= anatomical L5 in 5-vertebra Normal anatomy, where the two columns measure the same voxel set under different label schemes). Hip_m merges LH+RH; we train per-side and report both. TotalSegmentator numbers are Normal-subgroup zero-shot on CTSPINOPELVIC1K (Table 2); its combined Hips is placed in Hip_m. Top-block numbers are reproduced from those works' own test sets and are not directly comparable; bold marks best in column across all reported numbers. †Any-LSTV: case-weighted mean over the 7 LSTV test cases (per-subgroup numbers in Table 7).

Method	L1	L2	L3	L4	L5	last_lumbar	Sacrum	LH	RH	Hip_m
Reproduced from prior work; not evaluated on CTSPINOPELVIC1K *(Li et al. on COLONOG split, FracSegNet on PENGWIN test set)*										
3D U-Net	77.19	78.34	81.71	82.86	92.13	—	90.02	93.74	93.42	—
ResNet-50	80.29	76.41	80.62	83.77	91.05	—	88.37	94.72	94.01	—
ResNet-101	74.57	78.11	78.54	83.84	86.35	—	89.05	95.69	94.42	—
SwinUNETR	88.87	83.12	84.36	89.04	92.15	—	91.38	92.33	93.38	—
AttentionUNet	80.26	75.84	79.64	82.32	90.39	—	92.48	93.41	95.45	—
nnU-Net [Isensee et al., 2021]	89.35	91.37	92.41	93.37	95.74	—	96.72	95.50	96.78	—
Li et al. (CFFM+LT) [Li et al., 2025]	92.98	92.36	93.92	93.31	**97.07**	—	96.98	94.37	96.77	—
FracSegNet [Liu et al., 2025]	—	—	—	—	—	—	**98.7**	**99.3**	**99.2**	—
Reproducible baseline (publicly-released weights), evaluated zero-shot on CTSPINOPELVIC1K										
TotalSegmentator [Wasserthal et al., 2023]	92.1	91.8	92.9	93.6	93.8	—	81.7	—	—	92.9
Ours: CTSPINOPELVIC1K*, 5-fold ensemble on held-out test, partial-annotation supervision*										
Ours, Normal (n=167)	**95.3**	**95.5**	**95.9**	**96.7**	—	**97.1**	97.1	95.0	93.6	**98.8**
Ours, Any-LSTV (n=7)†	60.1	46.6	31.1	15.8	—	65.2	92.0	97.5	97.6	97.6

9.1 The Residual Class Collision

The merged scheme eliminates the L5/L6 collision driving TS's wrong-level mechanism. Each fold's checkpoint evaluates the same 174-case test split; voxel counts below pool 5 evaluations per case (15 case-fold pairs for lumb, 5 for sacr_count). Across 5-fold predictions on lumb cases ($\sim$4.06M aggregated GT-last_lumbar voxels), **58.2%** of GT-last_lumbar voxels are correctly predicted as last_lumbar (Block 1), against TS's structurally-zero last_lumbar emission. The residual 40.1% misallocation to L4 is the cost of label-merging at the joint, not a re-emergence of TS's full collapse. The merge does, however, expose a residual collision at the L4/last_lumbar boundary on sacr_count cases: GT-L4 voxels are predicted as last_lumbar in **82.7%** of voxels (Block 2; $\sim$720K GT-L4 voxels aggregated across 5 folds), and predicted-last_lumbar voxels on sacr_count cases are anatomically L4 in **100.0%** (Block 3). last_lumbar specificity on sacr_count = 0.778 ± 0.074 across folds.

Residual upper-lumbar caudal-shift cascade. While the merge restores correct labeling at the bottom-most boundary (Block 1), per-class DSC for upper lumbars on lumb cases shows a residual

Table 7: CTSPINOPELVIC1K 5-fold ensemble on held-out test split (n=174), reported as cross-fold mean ± std. LL EMIT: cases emitting ≥ 100 last_lumbar voxels. Dashes: classes without GT in that subtype. Semi-sacralization absent from test split (§10). Δ vs. TS zero-shot (Table 2). *n=1 subgroup: ± omitted (cross-fold std on a single case reflects fold-checkpoint disagreement on one outlier rather than a population distribution; sacr_count L4 in particular shows high cross-fold variability with mean 0.187 and per-fold dice spanning near-zero to mid-range).

Subgroup	n	**L1**	**L2**	**L3**	**L4**	**last_lumbar**	**Sac**	**Hip$_m$**	**Jxn DSC**	**LL emit**
Normal	167	.953±.008	.955±.005	.959±.004	.967±.005	.971±.005	.971±.002	.988±.001	.907±.016	167/167
Lumbarization	3	.872±.050	.611±.123	.377±.147	.213±.114	**.698±.038**	.946±.004	.960±.002	**.946±.007**	3/3
Sacralization (count)*	1	.988	.959	.734	.187	—	—	—	—	1/1
Sacralization (full)	2	.001±.002	.000±.000	.001±.002	.062±.083	.583±.136	.910±.039	.989±.000	.927±.068	2/2
Ambiguous*	1	—	—	—	—	—	.862	1.000	—	1/1
Δ vs. TS (lumb)	—	−0.6	−24.2	−31.3	−17.5	N/A	+11.3	+3.2	**+58.9**	—
Δ vs. TS (sacr_count)	—	+72.3	+85.9	+67.9	+18.6	—	—	—	—	—

Table 8: Residual class confusion under merged labels, focused on L4/last_lumbar. Voxel counts aggregated across 5 folds; percentages computed from the summed counts (denominator-weighted, not averaged percentages). Block 1: GT-last_lumbar on lumb (good signal, ∼4.06M voxels). Block 2: GT-L4 on sacr_count (failure-mode signal, ∼720K voxels). Block 3: pred-last_lumbar on sacr_count (inverse direction, ∼595K voxels).

Block	last_lumbar	sacrum	L4	background
Block 1: last_lumbar GT on lumb	**58.2%**	0.4%	40.1%	1.2%
Block 2: L4 GT on sacr_count	*82.7%*	0.3%	**14.9%**	2.2%
Block 3: pred-last_lumbar on sacr_count	—	0.0%	**100.0%**	0.0%

caudal-shift the merge does not resolve: L1 = 0.872, L2 = 0.611, L3 = 0.377, L4 = 0.213 (Table 7; std in table). The 5-vertebra positional prior anchors top-of-spine correctly but shifts labels caudally as the count discrepancy compounds. TS's shift concentrates at the bottom (no L6 class); ours distributes it through the upper lumbars (the merge absorbs the bottom shift into last_lumbar)—on per-class L2–L4, TS is in fact better, trading upper-lumbar precision for bottom-anchor correctness. Junction-DSC, centered on L5/S1, closes the surgically critical gap (+58.9 pts) regardless: wrong-level risk is dominated by the bottom-most label. Fully resolving the upper cascade requires a sequence predictor with explicit count constraints—contribution (iv) realized empirically: merge alone is necessary but insufficient.

Interpretation. The model has learned the spinopelvic spatial relationship on normal anatomy and recovers the merged last_lumbar class on lumbarization. On count-style sacralization, however, the positional prior dominates: the bottom-most lumbar visible in a given patch is treated as last_lumbar regardless of the global vertebral count. The disambiguating signal (vertebra count from a known anchor) is non-local; a 256×320×320 patch does not reliably contain both the caudal lumbosacral and rostral cervico-thoracic boundaries needed to count. This is a finding rather than a defect of the baseline—it executes the cleanest test of VERIDAH's training-side merge in the semantic-segmentation setting and confirms the architectural hypothesis underlying its two-stage design: the merge addresses class imbalance, but distinguishing the last-lumbar voxels of a 4-vertebra spine from those of a 5-vertebra spine must be deferred to a downstream component with access to global anatomical structure.

9.2 Where the Extra Vertebra Is: Eighteen Six-Lumbar Spines

Before the one-shot folds report, the released labels already answer the question the decoder is built for, on the 18 records that carry a sixth lumbar body (Fig. 6). Every measurement is a z-score against the 643 records with five rib-free bodies, no transitional flag and no hardware, so "normal" is the corpus's own distribution.

The bottom body. Four junction features signed toward the sacral type (a taller transverse process, a smaller process-to-ilium gap, a thinner disc beneath, a shorter sacrum) give a sacral-type score. Six records exceed 1.5 or carry a sacrum two standard deviations short; their processes sit 2–5 SD taller than a normal L5's and 2 SD closer to the ilium. Their Castellvi grades are IIa, IIb, IIb, IIb, IIIb

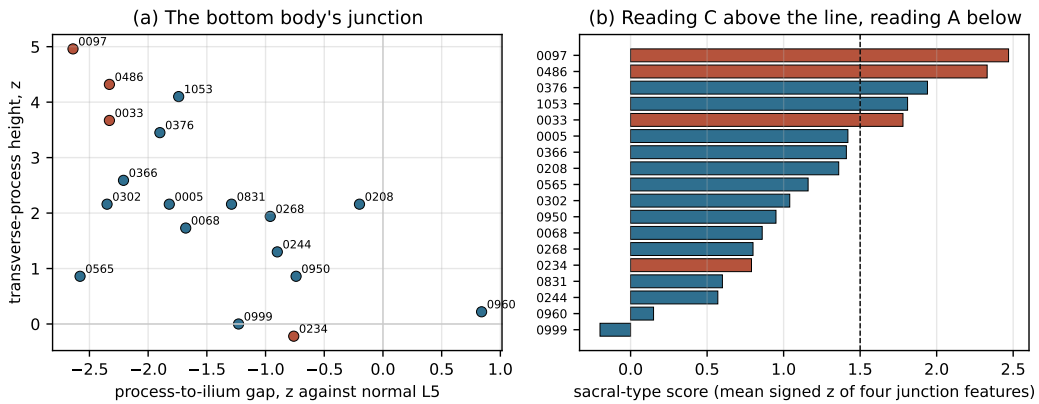


Figure 6: The bottom body of each six-lumbar record. (a) Transverse-process height against process-to-ilium gap, as z against normal L5. (b) The sacral-type score; records above the line read as a lumbarized S1. Colour is the final reading after the column-wide shift test.

and IV. Because a bilaterally fused process is carried inside the sacrum label, the process features go blind exactly at grade III, which is why sacral height is in the score.

The top body. Rib length cannot see an aplastic twelfth rib: the labels define the lowest rib-bearing vertebra as T12, so a spine whose true T12 has no ribs gets its T11 ribs called "rib 12" and every ratio looks normal. The test is therefore put to the bodies. Each level of each column is scored under per-level shape models fitted on the normal records, once as labelled and once with every name moved up one; on normal controls the as-labelled reading wins in 37 of 40 with a floor near -4 log-likelihood units. Four six-lumbar records fit better shifted up, their labelled L1 being a better T12 than L1.

Readings. 11 records read as a true L6, 4 as a lumbarized S1, 1 (an instrumented record) as an aplastic-rib T12, and 2 carry evidence at both ends, which is a real anatomy (a presacral count of 25 with a rib-less T12) that the field of view cannot exclude. That last pair is the case for a posterior rather than a verdict, and it is what the decoder reports.

Do the two borders shift together? Stump twelfth ribs (length below a third of the eleventh) occur in 98 of 788 records and lumbar ribs in 16 of 802. Among the 14 sacralizations, 4 carry a stump rib against 94 of 758 elsewhere (odds ratio 2.8, p=0.09); no lumbar rib occurs among the 14 lumbarizations. The direction agrees with Nagata et al. (2025); the power does not, at 33 graded records, and the whole-cohort Castellvi read that would settle it is in progress.

9.3 The One-Shot Folds

[PENDING: five folds launched 7 September 2026; expected complete about 12 September. To fill from `tools/eval_oneshot.py` and `tools/decode_sequence.py`: (i) per-class Dice on typical and on transitional records separately; (ii) rib-free count accuracy by transitional status and the calibration of the decoder's posterior; (iii) where ground-truth twelfth-rib, thirteenth-rib and lumbar-rib voxels went, with lumbar-rib and T13 recall; (iv) the decoder's readings on the 18 six-lumbar records against §9.2.]

10 Discussion and Future Work

Generalization and hardware. Any spine segmentation model lacking an L6 class will exhibit the level-shift on lumbarization; the LSTV gap is the symptom, not a model-specific bug. Our 100 GB patch budget requires H200-class hardware; smaller budgets yield weaker gap closure and therefore residual wrong-level risk downstream.

What the readings say about development. The 18 six-lumbar spines put the extra segment at the lumbosacral end (a true L6 or a lumbarized S1) far more often than at the thoracolumbar end, and the

two records with evidence at both ends are the anatomy a single homeotic shift of the whole column predicts [du Plessis et al., 2018, Poolman et al., 2023]. The stump-rib and sacralization co-occurrence points the same way at low power. VerSe [Sekuboyina et al., 2021], 460 volumes with an L6 label and no pelvis, is the cohort on which the one-shot network and the decoder can test the shift hypothesis at scale, once our sacral segmentation (DSC 0.969 ± 0.002, 5-fold) supplies the missing pelvis as a pseudolabel.

What it means at the table. Whether the bottom body is an L6 or a lumbarized S1 decides which junction is mobile, how large the pedicle is and where its trajectory runs, which nerve root exits beneath it, and which endplate the pelvic parameters were measured from. A segmenter that reports the type of the border body with a calibrated posterior gives the surgeon the one number the count cannot.

Limitations. COLONOG is a US screening cohort (ages 45–75); generalization to emergency/pediatric populations is not evaluated. LSTV labels are inherited from CTPelvic1K and CTSpine1K heuristics; Castellvi annotation covers all 33 LSTV cases but does not screen the `normal` cohort for Type I, and the source-dataset construction biases against morphologically-mild Type I presentations (§5). The held-out test split contains 7 of the 33 LSTV cases (3 lumb, 1 sacr_count, 2 sacralization-morph, 1 ambiguous, 0 semi-sacralization); fold-level evaluation pools across the same 174-case test set, yielding 15 case-fold pairs for lumbarization and 5 for sacr_count in voxel-confusion analysis but only the test-split n for case-level Dice (voxels within a case are highly correlated, so voxel counts in §9.1 represent dependent observations rather than independent samples). We do not run a same-compute unmerged 9-class baseline as a controlled ablation of the merge contribution; the merge's role is inferred from comparison to TS (which has no L6 class) rather than a same-data ablation. TS is evaluated zero-shot, the correct comparison for using TS as a production system but undersells what it could achieve if retrained with an L6 class.

Implications. A production CT segmentation system that emits a confident, locally well-formed, caudally-shifted prediction on a 5–35%-prevalence congenital variant produces no error signal at inference time and, when consumed downstream, reproduces the canonical mechanism of wrong-level spine surgery. Any future spine segmentation model claiming readiness for surgical-planning integration should report per-subgroup DSC and per-class confusion at the L5/S1 junction on a stratified LSTV cohort.

Data and Code Availability

CTSPINOPELVIC1K is hosted on HuggingFace Datasets at `https://huggingface.co/datasets/anonymous-neurips-ED/CTSpinoPelvic1K` with both core and Responsible AI fields of the Croissant specification [Akhtar et al., 2024]; a 50-record sample covering all six LSTV subtypes is available at `https://huggingface.co/datasets/anonymous-neurips-ED/CTSpinoPelvic1K-Sample`. The evaluation protocol, data-construction pipeline, and trainer source are released under permissive license at `https://github.com/anonymous-neurips-ED/spinopelvic-seg`, accessible without credentialed access. Trained 5-fold checkpoints will be made available. All repositories are anonymized for double-blind review.

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